

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Practical Lab Exam 1

Practical Lab Exam 1 - Answer Key

Class: Class 8

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Python Functions and Data Structures

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Lab Tasks (15 marks)

Task 1: Function-Based Calculator (5 marks)

Complete Python Program:

• Answer:

```
def add(a, b):
    """Return sum of two numbers."""
    return a + b

def subtract(a, b):
    """Return difference of two numbers."""
    return a - b

def multiply(a, b):
    """Return product of two numbers."""
    return a * b

def divide(a, b):
    """Return division of two numbers with zero check."""
    if b == 0:
        return "Error: Division by zero"
    return a / b

def main():
    while True:
        print("\n=== Calculator Menu ===")
        print("1. Add")
        print("2. Subtract")
        print("3. Multiply")
        print("4. Divide")
        print("5. Exit")

        choice = input("\nEnter your choice (1-5): ")

        if choice == '5':
            print("Thank you for using the calculator!")
            break

        if choice not in ['1', '2', '3', '4']:
            print("Invalid choice! Please try again.")
            continue

        try:
            num1 = float(input("Enter first number: "))
            num2 = float(input("Enter second number: "))
        except ValueError:
```

```

        print("Invalid input! Please enter numbers.")
        continue

    if choice == '1':
        result = add(num1, num2)
        print(f"{num1} + {num2} = {result}")
    elif choice == '2':
        result = subtract(num1, num2)
        print(f"{num1} - {num2} = {result}")
    elif choice == '3':
        result = multiply(num1, num2)
        print(f"{num1} × {num2} = {result}")
    elif choice == '4':
        result = divide(num1, num2)
        print(f"{num1} ÷ {num2} = {result}")

if __name__ == "__main__":
    main()

```

- **Marking:** 1 mark for function definitions, 1 mark for division by zero handling, 1 mark for menu-driven approach, 1 mark for input validation, 1 mark for formatted output.

Task 2: List Operations Manager (5 marks)

Complete Python Program:

- **Answer:**

```

def display_list(lst):
    """Display the current list."""
    if not lst:
        print("List is empty")
    else:
        print("Current list:", lst)

def main():
    names = ["Alice", "Bob", "Charlie", "Diana", "Eve",
             "Frank", "Grace", "Henry", "Ivy", "Jack",
             "Kate", "Leo", "Mia", "Noah", "Olivia"]

    while True:
        print("\n=== List Operations Manager ===")
        print("1. Add a name")
        print("2. Remove a name")
        print("3. Search for a name")
        print("4. Sort the list")
        print("5. Reverse the list")
        print("6. Display all names")
        print("7. Exit")

        choice = input("\nEnter your choice (1-7): ")

        if choice == '7':
            print("Goodbye!")
            break

        if choice == '1':
            name = input("Enter name to add: ")
            names.append(name)

```

```

        print(f"'{name}' added successfully!")
        display_list(names)

    elif choice == '2':
        name = input("Enter name to remove: ")
        if name in names:
            names.remove(name)
            print(f"'{name}' removed successfully!")
        else:
            print(f"Error: '{name}' not found in the list!")
            display_list(names)

    elif choice == '3':
        name = input("Enter name to search: ")
        if name in names:
            print(f"'{name}' found at index {names.index(name)}")
        else:
            print(f"'{name}' not found in the list")

    elif choice == '4':
        names.sort()
        print("List sorted alphabetically!")
        display_list(names)

    elif choice == '5':
        names.reverse()
        print("List reversed!")
        display_list(names)

    elif choice == '6':
        display_list(names)

    else:
        print("Invalid choice! Please try again.")

if __name__ == "__main__":
    main()

```

- **Marking:** 1 mark for list creation, 1 mark for menu implementation, 1 mark for list methods, 1 mark for input validation, 1 mark for display after operations.

Task 3: Dictionary-Based Marks Manager (5 marks)

Complete Python Program:

- **Answer:**

```

def calculate_average(marks):
    """Calculate average of marks."""
    return sum(marks.values()) / len(marks)

def find_topper(students):
    """Find student with highest average."""
    topper = None
    highest_avg = -1
    for name, marks in students.items():
        avg = calculate_average(marks)
        if avg > highest_avg:
            highest_avg = avg

```

```

        topper = name
    return topper, highest_avg

def display_students(students):
    """Display all students and their marks."""
    print(f"\n{'Name':<10} {'Math':<8} {'Science':<10} {'English':<10} {'SST':<8}  
{'Average':<10}")
    print("-" * 56)
    for name, marks in students.items():
        avg = calculate_average(marks)
        print(f"{'name':<10} {'marks['Math']':<8} {'marks['Science']':<10}  
{'marks['English']':<10} {'marks['SST']':<8} {'avg':<10.1f}")

def main():
    students = {
        "Alice": {"Math": 85, "Science": 90, "English": 78, "SST": 88},
        "Bob": {"Math": 70, "Science": 75, "English": 80, "SST": 65},
        "Charlie": {"Math": 95, "Science": 92, "English": 88, "SST": 90},
        "Diana": {"Math": 82, "Science": 78, "English": 85, "SST": 80},
        "Eve": {"Math": 88, "Science": 85, "English": 92, "SST": 78},
        "Frank": {"Math": 72, "Science": 80, "English": 75, "SST": 82},
        "Grace": {"Math": 90, "Science": 88, "English": 85, "SST": 92},
        "Henry": {"Math": 78, "Science": 82, "English": 80, "SST": 75}
    }

    while True:
        print("\n=== Marks Manager ===")
        print("1. Add new student")
        print("2. Update marks")
        print("3. Display all students")
        print("4. Calculate averages")
        print("5. Find topper")
        print("6. Exit")

        choice = input("\nEnter your choice (1-6): ")

        if choice == '6':
            print("Goodbye!")
            break

        if choice == '1':
            name = input("Enter student name: ")
            if name in students:
                print("Student already exists!")
            else:
                students[name] = {}
                for subject in ["Math", "Science", "English", "SST"]:
                    marks = int(input(f"Enter {subject} marks: "))
                    students[name][subject] = marks
                print(f"Student '{name}' added successfully!")

        elif choice == '2':
            name = input("Enter student name: ")
            if name not in students:
                print("Student not found!")
            else:

```

```

        subject = input("Enter subject (Math/Science/English/SST): ")
        if subject not in ["Math", "Science", "English", "SST"]:
            print("Invalid subject!")
        else:
            marks = int(input(f"Enter new {subject} marks: "))
            students[name][subject] = marks
            print("Marks updated successfully!")

    elif choice == '3':
        display_students(students)

    elif choice == '4':
        display_students(students)

    elif choice == '5':
        topper, avg = find_topper(students)
        print(f"\nTopper: {topper} with average {avg:.1f}")

    else:
        print("Invalid choice! Please try again.")

if __name__ == "__main__":
    main()

```

- **Marking:** 1 mark for dictionary structure, 1 mark for add/update functions, 1 mark for average calculation, 1 mark for topper finding, 1 mark for formatted output.

Viva Voce (5 marks)

1. **What is the difference between a function definition and a function call?**
 - **Answer:** Function definition is where you write the function code using def keyword. Function call is where you execute the function by using its name followed by parentheses.
 - **Marking:** 1 mark for correct explanation.
2. **Can a function have both default parameters and variable-length arguments?**
 - **Answer:** Yes. Example: `def func(a, b=5, *args):` where a is required, b has default value, and args accepts variable arguments.
 - **Marking:** 1 mark for correct answer with example.
3. **Why are dictionaries preferred over lists for key-value pairs?**
 - **Answer:** Dictionaries provide O(1) average time complexity for lookups using keys. Lists require O(n) time to search for values. Dictionaries also provide meaningful key names for better code readability.
 - **Marking:** 1 mark for explaining efficiency and readability.
4. **What is the difference between `remove()` and `pop()` for lists?**
 - **Answer:** `remove(x)` searches for value x and removes first occurrence, returns nothing. `pop(index)` removes and returns element at specified index, or last element if no index given.
 - **Marking:** 1 mark for explaining both methods.
5. **How does Python handle memory with deep copy vs shallow copy?**
 - **Answer:** Shallow copy creates new object but references same inner objects (changes affect original). Deep copy creates new object and recursively copies all inner objects (completely independent). Use `copy.deepcopy()` for nested structures.
 - **Marking:** 1 mark for explaining memory handling difference.